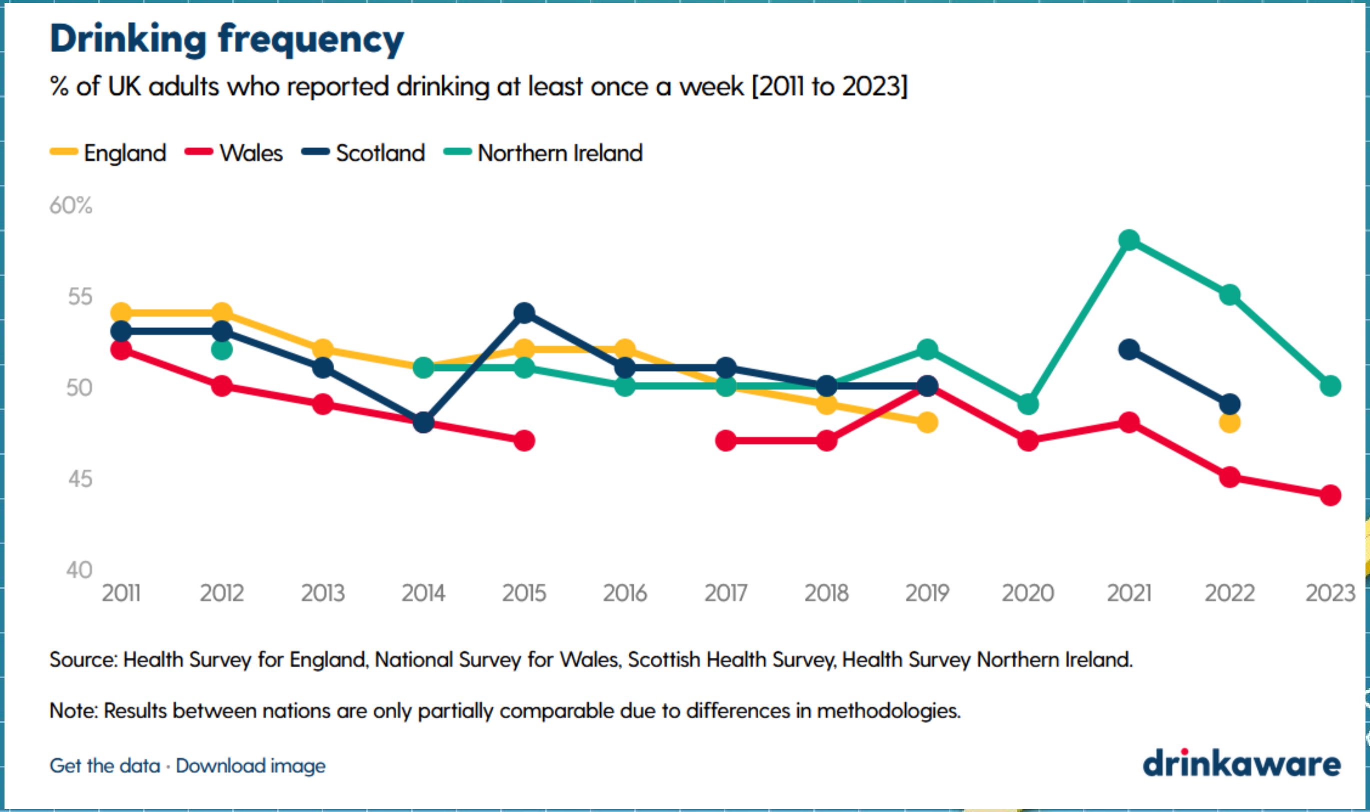
Statistical Analysis

Alcohol in the United Kingdom
(Health Survey for England, 2011)

Presentation overview

- Context: alcohol in the UK, a major issue
- The data: descriptive statistics
- Inferential statistics
- Interpretations and recommendations
- Appendices

Introduction



Health Survey for England, 2022 Part 1 - NHS England Digital

The proportion of children aged 8 to 15 who had ever drunk alcohol fell from 45% in 2003 to 14% in 2022.

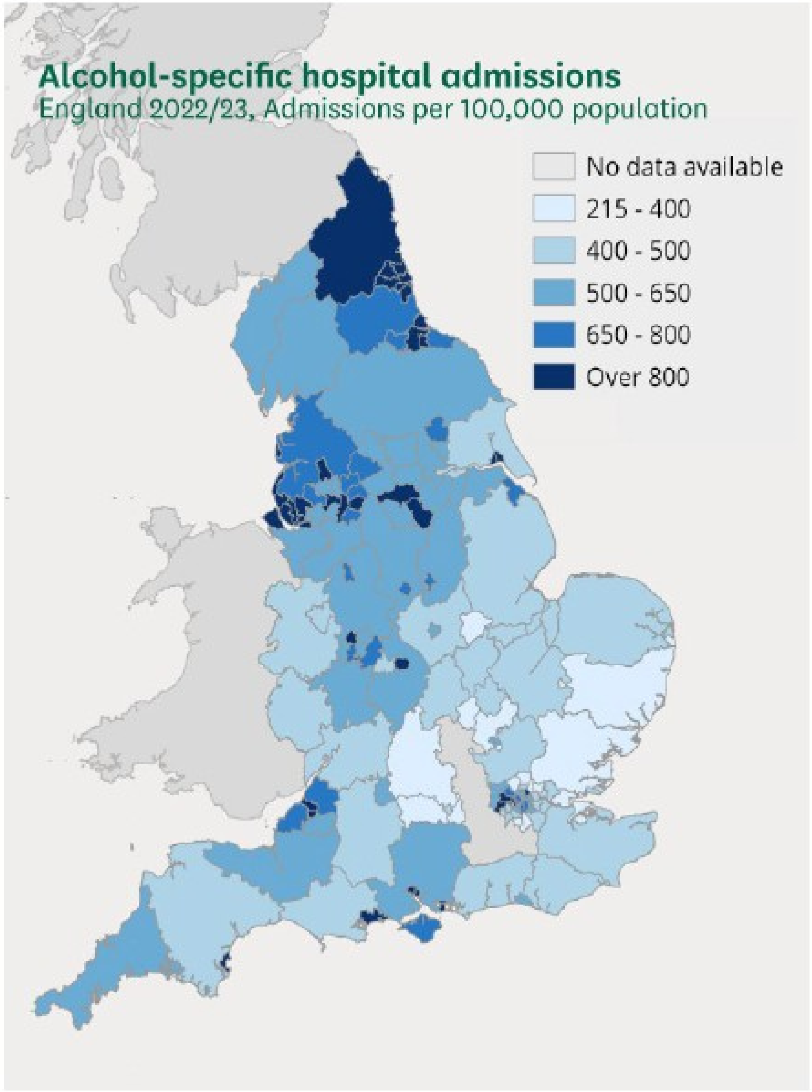
A higher proportion of men (32%) than women (15%) drank at levels that put them at increasing or higher risk of alcohol-related harm (over 14 units in the last week).

In 2022, more men (84%) than women (78%) had drunk alcohol in the last 12 months.

Hospital admissions
per 100,000

10 HIGHEST RATES	
Southampton	1,981
South Tyneside	1,260
Blackpool	1,219
North Tyneside	1,208
Hartlepool	1,183
Newcastle upon Tyne	1,152
Salford	1,128
Bristol	1,111
Gateshead	1,105
Sunderland	1,082

10 LOWEST RATES	
Rutland	220
Wokingham	280
Redbridge	306
Essex	315
Sutton	320
Enfield	323
Thurrock	358
West Berkshire	360
Central Bedfordshire	363
Barnet	365



Source: PHE Local Alcohol Profiles

Commons Library Research Briefing, 17 July 2024

The data

10,617

63.22%

54.30%

Degree or equivalent

7.70%

Descriptive statistics of household size, BMI, and age

Household Size:

- Mean: 2.85
- Median: 3
- Mode: 2
- Minimum: 1
- Maximum: 10
- Range: 9
- Standard Deviation: 1.37

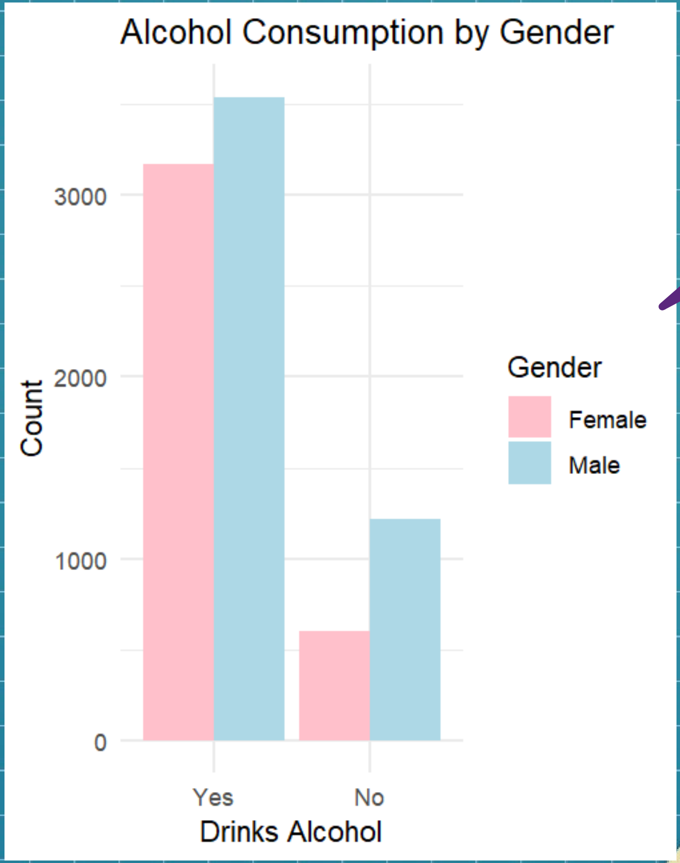
BMI:

- Mean: 25.92
- Median: 25.59
- Mode: 25.32
- Minimum: 8.34
- Maximum: 65.28
- Range: 56.94
- Standard Deviation: 6.14

Age

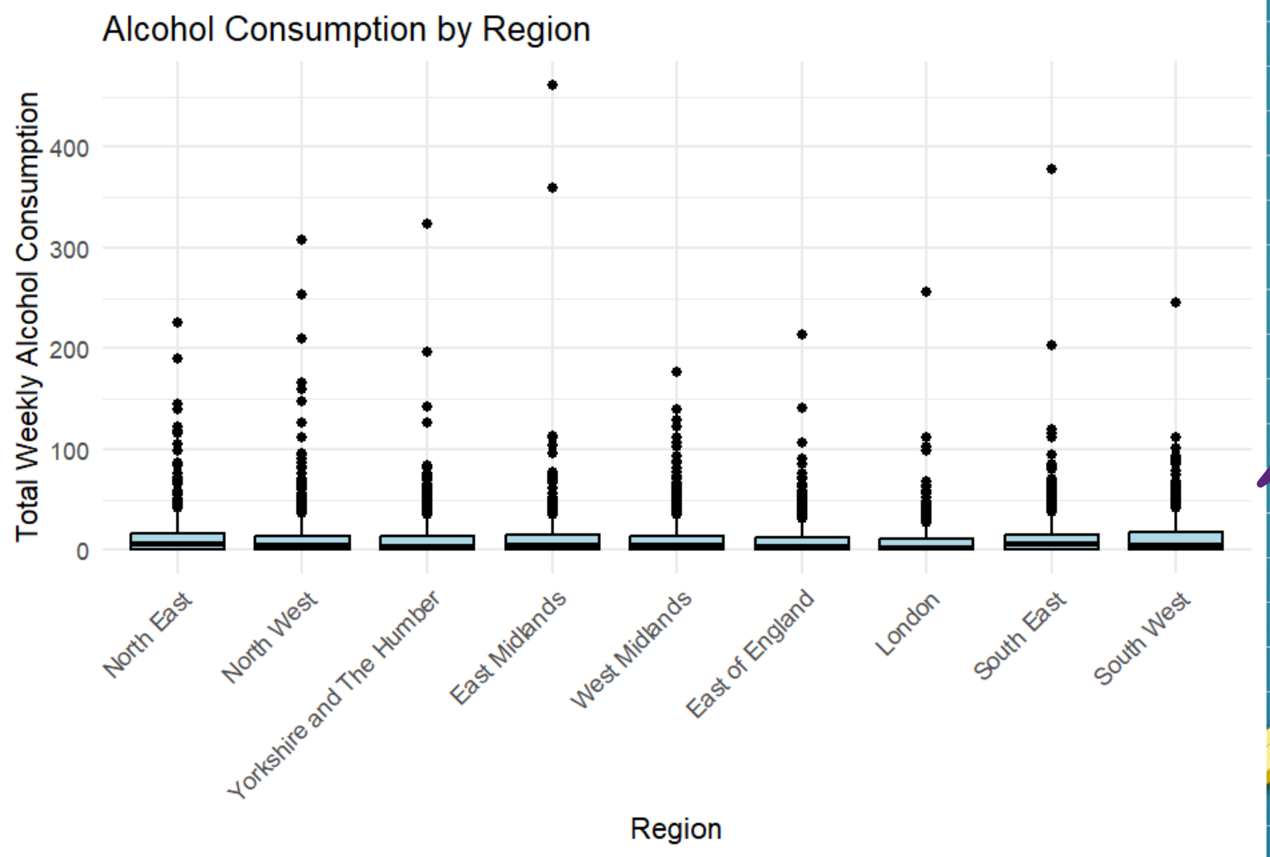
- Mean: 41.56
- Median: 42
- Mode: 64
- Minimum: 0
- Maximum: 100
- Range: 100
- Standard Deviation: 23.83

Inferential statistics (1)



Chi-Square Test

Inferential statistics (2)



ANOVA (Analysis of Variance) test

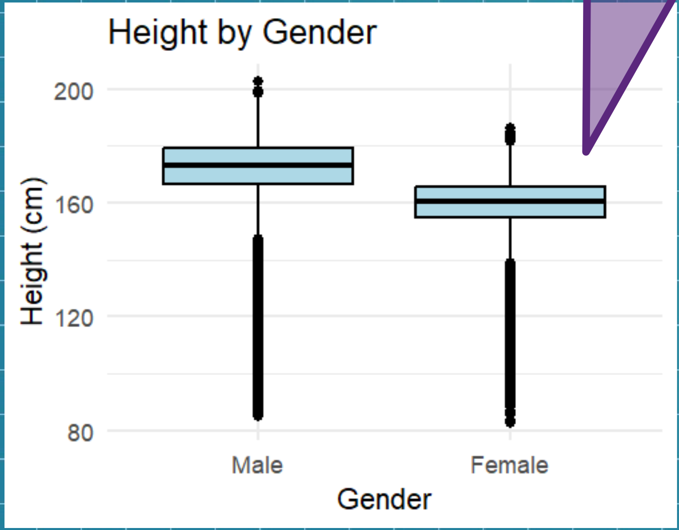
Inferential statistics (2)

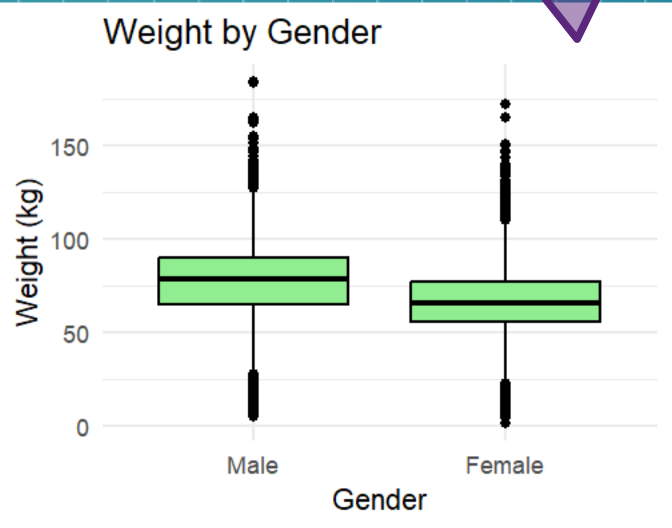
Tukey HSD Test

Inferential statistics (3)

T-Test

T-Test

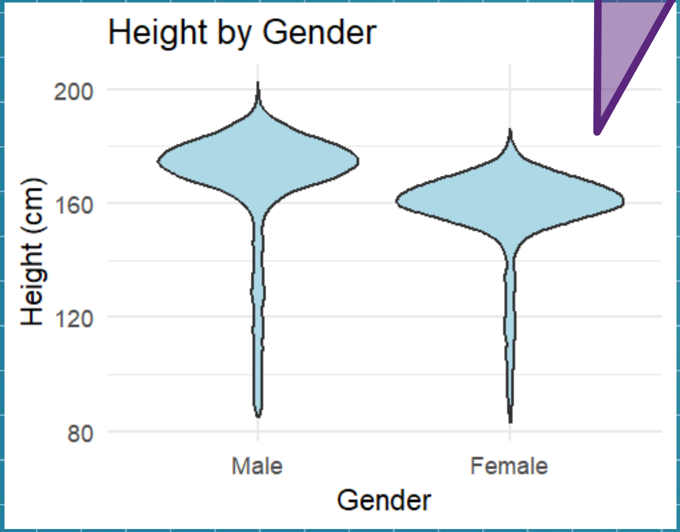


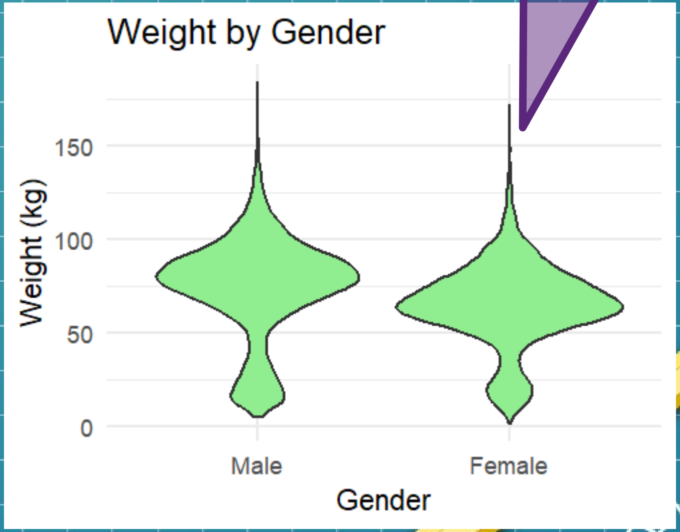


Inferential statistics (3)

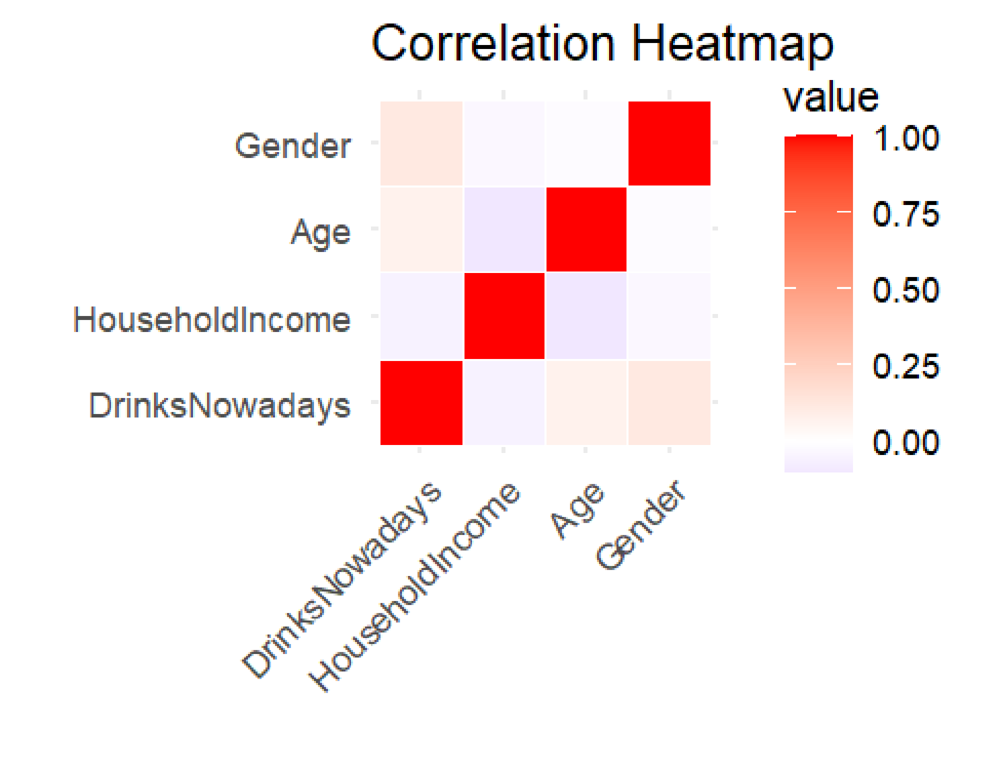
T-Test

T-Test

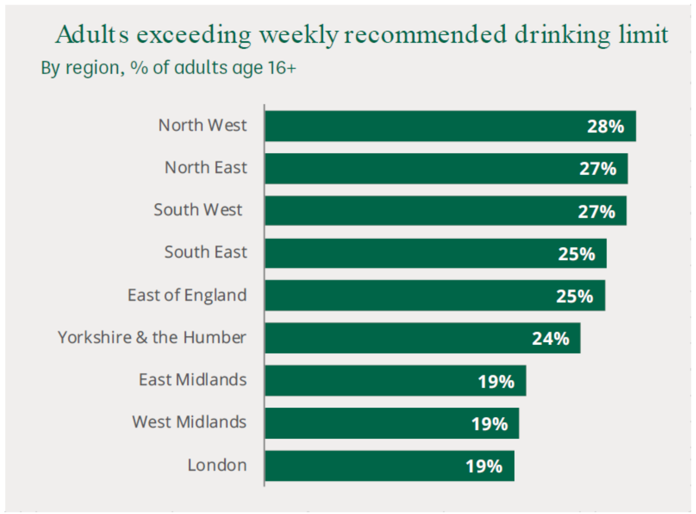




Inferential statistics (4)

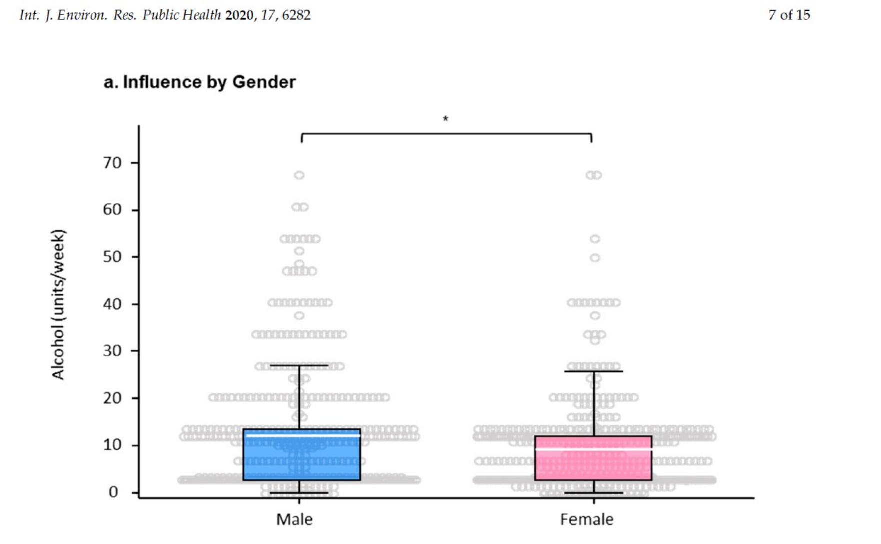


Findings and current statistics



Source: Health Survey for England 2022, Adult drinking

Findings and current statistics



Notes: The Health Survey for England was suspended in 2020 due to the coronavirus pandemic. The dotted area of the chart highlights this gap in the data series.

Source: Health Survey for England 2022, Adult drinking

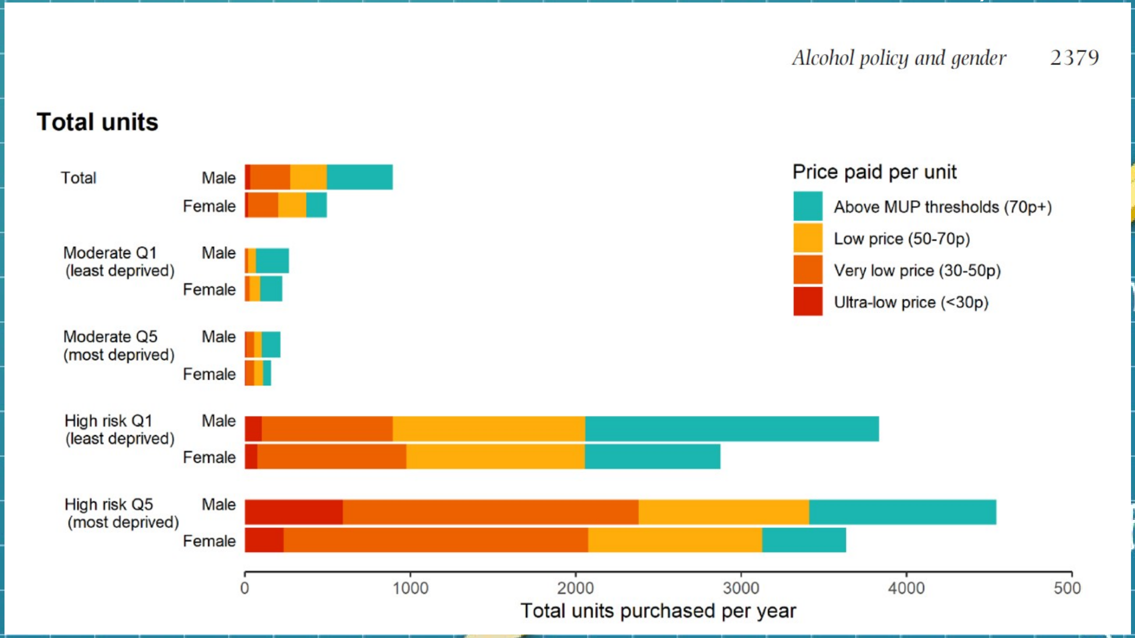
Conclusions and recommendations

From Petra S. Meier & ali., 2020

- Targeted interventions : educating high-risk groups

- Differentiate the way to address both genders, especially in university and in deprived areas

- Long-term cultural shifts



Thank you

Sources

- Alcohol Change UK (2024) Drinking trends in the UK. Available from: https://alcoholchange.org.uk/alcohol-facts/fact-sheets/drinking-trends-in-the-uk [Accessed 18 October 2024].

- Bhatti, S.N., Fan, L.M., Collins, A. & Li, J.-M. (2020) Exploration of Alcohol Consumption Behaviours and Health-Related Influencing Factors of Young Adults in the UK. *International Journal of Environmental Research and Public Health*. 17 (17), 6282. Available from: doi:10.3390/ijerph17176282 [Accessed 18 October 2024].

- Drinkaware (2024) Alcohol consumption UK. Available from: https://www.drinkaware.co.uk/research/alcohol-facts-and-data/alcohol-consumption-uk [Accessed 18 October 2024].

- Health Survey for England (2022) *Health Survey for England, 2022 Part 1*. National statistics. Available from: https://digital.nhs.uk/data-and-information/publications/statistical/health-survey-for-england/2022-part-1/adult-drinking [Accessed 18 October 2024].

- Meier, P.S., Holmes, J., Brennan, A. & Angus, C. (2021) Alcohol policy and gender: a modelling study estimating gender-specific effects of alcohol pricing policies. *Addiction*. 116 (9), 2372–2384. Available from: doi:10.1111/add.15464 [Accessed 18 October 2024].

- Stiebahl, S., (2024) Alcohol statistics: England. Commons Library Research Briefing, *House of Commons Library*, 17 July. Available from: commonslibrary.parliament.uk [Accessed 18 October 2024].

Appendices

```
> #loading the dataset
>
> library(haven)
>
> data <- read_sav("C:/Users/Marie LEVESQUE/OneDrive/Documents/ESSEX_Msc_Da
taScience/3_Numerical_statistical_analysis/unit_4_to_7_mathematics_test_pre
p/Assignment/HSE 2011.sav")
>
> head(data)
# A tibble: 6 × 58
  hserial  pserial HHSize tenureb Sex        Age MonthAge WeekAge    PersNo
    <dbl>    <dbl>  <dbl> <dbl+l> <dbl+l>  <dbl>    <dbl> <dbl+lbl>   <dbl>
1 1001011   1.00e8      1 1 [Own… 2 [Fem…     75       12 997 [Ove…       1
2 1001031   1.00e8      3 1 [Own… 2 [Fem…     47       12 997 [Ove…       1
3 1001041   1.00e8      2 1 [Own… 1 [Mal…     77       12 997 [Ove…       1
4 1001041   1.00e8      2 1 [Own… 2 [Fem…     66       12 997 [Ove…       2
5 1001051   1.00e8      1 1 [Own… 1 [Mal…     44       12 997 [Ove…       1
6 1001061   1.00e8      2 1 [Own… 1 [Mal…     66       12 997 [Ove…       1
# ℹ 49 more variables: topqual3 <dbl+lbl>, HRPID <dbl+lbl>,
    conact <dbl+lbl>, nssec8 <dbl+lbl>, Origin <dbl+lbl>,
    otinc <dbl+lbl>, eqvinc <dbl+lbl>, NurOutc <dbl+lbl>,
    elto01 <dbl+lbl>, relto02 <dbl+lbl>, relto03 <dbl+lbl>,
    elto04 <dbl+lbl>, relto05 <dbl+lbl>, relto06 <dbl+lbl>,
    elto07 <dbl+lbl>, relto08 <dbl+lbl>, relto09 <dbl+lbl>,
    elto10 <dbl+lbl>, Relto11 <dbl+lbl>, Relto12 <dbl+lbl>, …
```

R code for loading the data / converting into csv to have a first look at it

```
> library(haven)
> alcohol_uk <- read_sav("C:/Users/Marie LEVESQUE/OneDrive/Documents/ESSEX_Msc_Da
taScience/3_Numerical_statistical_analysis/unit_4_to_7_mathematics_test_prep/Assi
gnment/HSE 2011.sav")
>
> write.table(alcohol_uk, file = "C:/Users/Marie LEVESQUE/OneDrive/Documents/ESSE
X_Msc_DataScience/3_Numerical_statistical_analysis/unit_4_to_7_mathematics_test_p
rep/Assignment/from_sav_data.csv", quote = FALSE, sep = ",")
>
```

```
> # Check unique values in the columns of interest
> columns_to_check <- c("topqual3", "marstatc", "dnnow", "totalwu", "porfv")
>
> # Loop through columns and check unique values
> for (col in columns_to_check) {
+     print(paste("Unique values in column:", col))
+     print(unique(data[[col]]))
+ }
```

Checking for the labels associated with region, marital status and education reference values
The code has been used to find alcohol related columns as well and understand the abbreviated labels

```
[1] "unique values in column: gor1"
> print(unique(data[[col]]))
<labelled<double>[9]>: Government Office Region - numeric
[1] 6 2 3 1 9 7 5 8 4

Labels:
 value                    label
     1               North East
     2               North West
     3 Yorkshire and The Humber
     4            East Midlands
     5            West Midlands
     6          East of England
     7                   London
     8               South East
     9               South West
    10                    Wales
    11                 Scotland
```

```
[1] "Unique values in column: marstatc"
<labelled<double>[8]>: (D) Marital status including cohabitees
[1]  5  2  1  6 NA  7  4  3

Labels:
 value                                         label
    -9                                       Refused
    -8                                    Don't know
    -7                          Refused/not obtained
    -6                          Schedule not obtained
    -2                         Schedule not applicable
    -1                                 Not applicable
     1                                        Single
     2                                       Married
     3 Civil partnership including spontaneous answers
     4                                     Separated
     5                                      Divorced
     6                                       Widowed
     7                                    Cohabitees
```

```
[1] "Unique values in column: topqual3"
<labelled<double>[8]>: (D) Highest Educational Qualification
[1]  6  4  1  3  7  2 NA  5

Labels:
 value                       label
    -9                     Refused
    -8                  Don't know
    -7        Refused/not obtained
    -6       Schedule not obtained
    -2     Schedule not applicable
    -1              Not applicable
     1   NVQ4/NVQ5/Degree or equiv
     2      Higher ed below degree
     3        NVQ3/GCE A Level equiv
     4        NVQ2/GCE O Level equiv
     5 NVQ1/CSE other grade equiv
     6               Foreign/other
     7             No qualification
```

```
> # Check unique values in the columns of interest
> columns_to_check <- c("omdiaval", "omsysval", "dnnow", "totalwu", "porfv")
>
> # Loop through columns and check unique values
> for (col in columns_to_check) {
+     print(paste("Unique values in column:", col))
+     print(unique(data[[col]]))
+ }
```

Looking for the columns that reference alcohol drinking : ddnow and totalwu

```
[1] "Unique values in column: dnnow"
<labelled<double>[3]>: Whether drink nowadays
[1]  2  1 NA

Labels:
 value                label
    -9              Refusal
    -8           Don't know
    -1 Item not applicable
     1                  Yes
     2                   No
[1] "Unique values in column: totalwu"
<labelled<double>[2665]>: (D) Total units of alcohol/week
   [1]   0.05800   4.99100  49.02900   0.00000  30.23000  13.55800  24.61500   4.61600
   [9]  47.75000   0.86600   4.68850        NA   4.50000   0.02900  10.98900   6.11600
```

```
> # Number of people in the sample
> total_people <- nrow(data)
>
> # Percentage of people who drink alcohol (1 in the dnnow "drinking now" column)
> alcohol_consumers <- sum(data$dnnow == 1, na.rm = TRUE)
> percentage_alcohol <- (alcohol_consumers / total_people) * 100
>
> # Percentage of women (1 = male, 2 = female in 'Sex' column)
> percentage_women <- (sum(data$Sex == 2, na.rm = TRUE) / total_people) * 100
>
> # Highest educational level (topqual3 is the education level column)
> highest_education_level <- min(data$topqual3, na.rm = TRUE)
>
> # Percentage of divorced and separated people (4=separated, 5=divorced in 'marstatc' column)
> divorced_separated <- sum(data$marstatc %in% c(4, 5), na.rm = TRUE)
> percentage_divorced_separated <- (divorced_separated / total_people) * 100
>
> # Summary statistics for household size (HHSize), BMI (bmival), and age (Age)
> household_size_stats <- summary(data$HHSize)
> age_stats <- summary(data$Age)
> bmi_stats <- summary(data$bmival)
>
> # Standard deviation for household size, age, and BMI
> sd_household_size <- sd(data$HHSize, na.rm = TRUE)
> sd_age <- sd(data$Age, na.rm = TRUE)
> sd_bmi <- sd(data$bmival, na.rm = TRUE)
>
> # Descriptive statistics for total weekly alcohol consumption (totalwu)
> totalwu_stats <- summary(data$totalwu)
> sd_totalwu <- sd(data$totalwu, na.rm = TRUE)
>
```

Descriptive statistics

Data	
data	10617 obs. of 58 variables
Values	
age_stats	'summaryDefault' Named num [1:6] 0 22 42 41.6 61 .
alcohol_consumers	6712L
bmi_stats	'summaryDefault' Named num [1:7] 8.34 21.93 25.59
divorced_separated	818L
highest_education_level	dbl+lbl [1:1] 1
household_size_stats	'summaryDefault' Named num [1:6] 1 2 3 2.85 4 ...
percentage_alcohol	63.2193651690685
percentage_divorced_se…	7.70462465856645
percentage_women	54.2997080154469
sd_age	23.8320278051991
sd_bmi	6.13834351764461
sd_household_size	1.36852803541081
sd_totalwu	20.0254574675987
total_people	10617L
totalwu_stats	'summaryDefault' Named num [1:7] 0 0.188 3.837 10.

Mean, median, mode, min, max, range, standard deviation (Household size, BMI, Age)

```
> # Function to calculate mode
> calc_mode <- function(x) {
+     uniq_x <- unique(na.omit(x))
+     uniq_x[which.max(tabulate(match(x, uniq_x)))]
+ }
>
> # Household Size (HHSize)
> mean_HHSize <- mean(data$HHSize, na.rm = TRUE)
> median_HHSize <- median(data$HHSize, na.rm = TRUE)
> mode_HHSize <- calc_mode(data$HHSize)
> min_HHSize <- min(data$HHSize, na.rm = TRUE)
> max_HHSize <- max(data$HHSize, na.rm = TRUE)
> range_HHSize <- max_HHSize - min_HHSize
> sd_HHSize <- sd(data$HHSize, na.rm = TRUE)
>
> # BMI (bmival)
> mean_BMI <- mean(data$bmival, na.rm = TRUE)
> median_BMI <- median(data$bmival, na.rm = TRUE)
> mode_BMI <- calc_mode(data$bmival)
> min_BMI <- min(data$bmival, na.rm = TRUE)
> max_BMI <- max(data$bmival, na.rm = TRUE)
> range_BMI <- max_BMI - min_BMI
> sd_BMI <- sd(data$bmival, na.rm = TRUE)
>
> # Age (Age at last birthday)
> mean_Age <- mean(data$Age, na.rm = TRUE)
> median_Age <- median(data$Age, na.rm = TRUE)
> mode_Age <- calc_mode(data$Age)
> min_Age <- min(data$Age, na.rm = TRUE)
> max_Age <- max(data$Age, na.rm = TRUE)
> range_Age <- max_Age - min_Age
> sd_Age <- sd(data$Age, na.rm = TRUE)
>
```

```
> # Output the results
> cat("Household Size Statistics:\n")
Household Size Statistics:
> cat("Mean:", mean_HHSize, "\n")
Mean: 2.850711
> cat("Median:", median_HHSize, "\n")
Median: 3
> cat("Mode:", mode_HHSize, "\n")
Mode: 2
> cat("Min:", min_HHSize, "\n")
Min: 1
> cat("Max:", max_HHSize, "\n")
Max: 10
> cat("Range:", range_HHSize, "\n")
Range: 9
> cat("Standard Deviation:", sd_HHSize, "\n\n")
Standard Deviation: 1.368528

>
> cat("BMI Statistics:\n")
BMI Statistics:
> cat("Mean:", mean_BMI, "\n")
Mean: 25.9205
> cat("Median:", median_BMI, "\n")
Median: 25.59321
> cat("Mode:", mode_BMI, "\n")
Mode: 25.32541
> cat("Min:", min_BMI, "\n")
Min: 8.34011
> cat("Max:", max_BMI, "\n")
Max: 65.27721
> cat("Range:", range_BMI, "\n")
Range: 56.9371
> cat("Standard Deviation:", sd_BMI, "\n\n")
Standard Deviation: 6.138344
```

```
>
> cat("Age Statistics:\n")
Age Statistics:
> cat("Mean:", mean_Age, "\n")
Mean: 41.56136
> cat("Median:", median_Age, "\n")
Median: 42
> cat("Mode:", mode_Age, "\n")
Mode: 64
> cat("Min:", min_Age, "\n")
Min: 0
> cat("Max:", max_Age, "\n")
Max: 100
> cat("Range:", range_Age, "\n")
Range: 100
> cat("Standard Deviation:", sd_Age, "\n")
Standard Deviation: 23.83203
```

```
> # Chi-Square test for gender and whether they drink alcohol
> table_gender_alcohol <- table(data$Sex, data$dnnow)
> chi_test_gender_alcohol <- chisq.test(table_gender_alcohol)
>
> # Output results
> cat("Chi-Square test results for gender and alcohol consumption:\n")
Chi-Square test results for gender and alcohol consumption:
> print(chi_test_gender_alcohol)

        Pearson's Chi-squared test with Yates' continuity
        correction

data:  table_gender_alcohol
X-squared = 114.15, df = 1, p-value < 2.2e-16
```

Chi-square test

chi_test_gender_alcohol	list [9] (S3: htest)	List of length 9
statistic	double [1]	114.1497
parameter	integer [1]	1
p.value	double [1]	1.20834e-26
method	character [1]	'Pearson\'s Chi-squared test with Yates\' continuity correction'
data.name	character [1]	'table_gender_alcohol'
observed	integer [2 x 2] (S3: table)	3172 3540 605 1217
expected	double [2 x 2]	2971 3741 806 1016
residuals	double [2 x 2] (S3: table)	3.69 -3.29 -7.09 6.32
stdres	double [2 x 2] (S3: table)	10.7 -10.7 -10.7 10.7

ANOVA Test

```
> # ANOVA to compare alcohol consumption across regions
> anova_region_alcohol <- aov(data$totalwu ~ as.factor(data$gor1))
>
> # Output ANOVA results
> cat("ANOVA test results for region and alcohol consumption:\n")
ANOVA test results for region and alcohol consumption:
> summary(anova_region_alcohol)
                       Df  Sum Sq Mean Sq F value   Pr(>F)    
as.factor(data$gor1)    8   14438  1804.8   4.515 1.71e-05 ***
Residuals            8475 3387405   399.7                     
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
2133 observations effacées parce que manquantes
>
```

anova_region_alcohol	list [14] (S3: aov, lm)	List of length 14
coefficients	double [9]	12.95 -1.76 -1.80 -1.12 -1.77 -3.38 ...
residuals	double [8484] (S3: haven_labelle	-9.52 -4.58 39.46 -9.57 20.66 3.98 ...
effects	double [8484] (S3: haven_labelle	-1004.79 9.80 8.88 29.29 15.38 35.16 ...
rank	integer [1]	9
fitted.values	double [8484]	9.57 9.57 9.57 9.57 9.57 9.57 ...
assign	integer [9]	0 1 1 1 1 1 ...
qr	list [5] (S3: qr)	List of length 5
df.residual	integer [1]	8475
na.action	integer [2133] (S3: omit)	13 14 40 41 43 61 ...
contrasts	list [1]	List of length 1
xlevels	list [1]	List of length 1
call	language	aov(formula = data$totalwu ~ as.factor(data$gor1))
terms	formula	data$totalwu ~ as.factor(data$gor1)
model	list [8484 x 2] (S3: data.frame)	A data.frame with 8484 rows and 2 columns

Tukey HSD Test

```
> # Output the results
> cat("Tukey HSD post-hoc test results:\n")
Tukey HSD post-hoc test results:
> print(tukey_test)
  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = data$totalwu ~ as.factor(data$gor1))

$`as.factor(data$gor1)`
            diff        lwr        upr     p adj
2-1 -1.764176129 -4.7510697  1.2227175 0.6604033
3-1 -1.800695614 -4.9269157  1.3255245 0.6909460
4-1 -1.119266078 -4.3576046  2.1190724 0.9781414
5-1 -1.774150625 -4.9019559  1.3536546 0.7088383
6-1 -3.375167351 -6.4658899 -0.2844448 0.0202781
7-1 -4.944499152 -8.0041116 -1.8848867 0.0000192
8-1 -1.733217716 -4.5976874  1.1312520 0.6296040
9-1 -1.186804964 -4.3170000  1.9433901 0.9615300
3-2 -0.036519485 -2.8423041  2.7692651 1.0000000
4-2  0.644910051 -2.2852796  3.5750997 0.9989976
5-2 -0.009974496 -2.8175252  2.7975762 1.0000000
6-2 -1.610991221 -4.3771693  1.1551868 0.6777131
7-2 -3.180323023 -5.9116970 -0.4489490 0.0092764
8-2  0.030958414 -2.4798904  2.5418072 1.0000000
9-2  0.577371165 -2.2328417  3.3875840 0.9993938
4-3  0.681429536 -2.3906594  3.7535184 0.9989393
5-3  0.026544989 -2.9287993  2.9818892 1.0000000
6-3 -1.574471736 -4.4905409  1.3415975 0.7619367
7-3 -3.143803538 -6.0268787 -0.2607284 0.0206168
8-3  0.067477899 -2.6076072  2.7425630 1.0000000
```

T-tests

```
> # t-test for height by gender
> t_test_height <- t.test(data$htval ~ data$Sex)
>
> # t-test for weight by gender
> t_test_weight <- t.test(data$wtval ~ data$Sex)
>
```

```
> # Output t-test results for height and weight
> cat("T-test results for height by gender:\n")
T-test results for height by gender:
> print(t_test_height)

        Welch Two Sample t-test

data:  data$htval by data$Sex
t = 25.197, df = 6964.1, p-value < 2.2e-16
alternative hypothesis: true difference in means between group 1 and group 2 is not
equal to 0
95 percent confidence interval:
  9.394798 10.979901
sample estimates:
mean in group 1 mean in group 2
       167.3928        157.2054

>
> cat("T-test results for weight by gender:\n")
T-test results for weight by gender:
> print(t_test_weight)

        Welch Two Sample t-test

data:  data$wtval by data$Sex
t = 17.847, df = 7777, p-value < 2.2e-16
alternative hypothesis: true difference in means between group 1 and group 2 is not
equal to 0
95 percent confidence interval:
  8.463743 10.552436
sample estimates:
mean in group 1 mean in group 2
       74.26612        64.75803
```

t_test_height × t_test_weight ×

Show Attributes

Name	Type	Value
t_test_height	list [10] (S3: htest)	List of length 10
statistic	double [1]	25.19749
parameter	double [1]	6964.05
p.value	double [1]	3.769636e-134
conf.int	double [2]	9.39 10.98
estimate	double [2]	167 157
null.value	double [1]	0
stderr	double [1]	0.4043002
alternative	character [1]	'two.sided'
method	character [1]	'Welch Two Sample t-test'
data.name	character [1]	'data$htval by data$Sex'

t_test_height × t_test_weight ×

Show Attributes

Name	Type	Value
t_test_weight	list [10] (S3: htest)	List of length 10
statistic	double [1]	17.84696
parameter	double [1]	7777.02
p.value	double [1]	7.449037e-70
conf.int	double [2]	8.46 10.55
estimate	double [2]	74.3 64.8
null.value	double [1]	0
stderr	double [1]	0.5327567
alternative	character [1]	'two.sided'
method	character [1]	'Welch Two Sample t-test'
data.name	character [1]	'data$wtval by data$Sex'

Correlation test

```
> # Convert categorical variables to numeric (e.g., 1 for male, 2 for female)
> data$Sex_numeric <- as.numeric(data$Sex)
>
> # Create a data frame for correlation analysis
> correlation_data <- data.frame(
+     DrinksNowadays = data$dnnow,
+     HouseholdIncome = data$totinc,
+     Age = data$Age,
+     Gender = data$Sex_numeric
+ )
>
> # Remove rows with missing values for the correlation analysis
> correlation_data <- na.omit(correlation_data)
>
> # Calculate correlation matrix using Spearman correlation
> correlation_matrix <- cor(correlation_data, method = "spearman")
>
> # Output correlation matrix
> cat("Correlation matrix for alcohol, income, age, and gender:\n")
Correlation matrix for alcohol, income, age, and gender:
> print(correlation_matrix)
                DrinksNowadays HouseholdIncome         Age      Gender
DrinksNowadays      1.00000000     -0.05695969  0.06837817  0.11668897
HouseholdIncome    -0.05695969      1.00000000 -0.10294637 -0.03321667
Age                 0.06837817     -0.10294637  1.00000000 -0.01733726
Gender              0.11668897     -0.03321667 -0.01733726  1.00000000
>
> 
```

Visualisation code (R)

Chi-square tests results (gender & alcohol)

```
> # Filter out rows where dnnow is NA
> data_clean <- data[!is.na(data$dnnow), ]
>
> # Bar plot of alcohol consumption by gender, excluding NAs
> ggplot(data_clean, aes(x = as.factor(dnnow), fill = as.factor(Sex))) +
+     geom_bar(position = "dodge") +
+     scale_fill_manual(values = c("pink", "lightblue"), labels = c("Female", "Male")) +
+     labs(x = "Drinks Alcohol", fill = "Gender", y = "Count", title = "Alcohol Consumption by Gender") +
+     scale_x_discrete(labels = c("1" = "Yes", "2" = "No")) +  # Map 1 to Yes, 2 to No on the x-axis
+     theme_minimal()
```

BoxPlot for alcohol consumption across regions
Plot for Tukey HSD test results

```
> # Remove non-finite values before plotting
> data_clean <- data[!is.na(data$totalwu) & is.finite(data$totalwu), ]
> # Create a vector of region labels
> region_labels <- c("North East", "North West", "Yorkshire and The Humber",
+                    "East Midlands", "West Midlands", "East of England",
+                    "London", "South East", "South West", "Wales", "Scotland")
>
> # Convert gor1 to a factor with region labels
> data_clean$region <- factor(data_clean$gor1,
+                             levels = 1:11,  # Match the levels to the unique values in gor1
+                             labels = region_labels)
>
> # Plot using the region labels
> ggplot(data_clean, aes(x = region, y = totalwu)) +
+     geom_boxplot(fill = "lightblue", color = "black") +
+     labs(x = "Region", y = "Total Weekly Alcohol Consumption", title = "Alcohol Consumption by Region") +
+     theme_minimal() +
+     theme(axis.text.x = element_text(angle = 45, hjust = 1))  # Rotate x-axis labels for readability
>
>
> # Plot Tukey HSD results
> plot(tukey_test, las = 1)  # las = 1 ensures labels are horizontal
>
> # Compact Letter Display (CLD) visualization
> tukey_cld <- multcompLetters4(anova_region_alcohol, tukey_test)
> print(tukey_cld)
```

Boxplot for T-tests

```
> # Filter out rows where htval is non-finite (e.g., NA, Inf)
> data_clean_height <- data[!is.na(data$htval) & is.finite(data$htval), ]
>
> # Boxplot for height by gender
> ggplot(data_clean_height, aes(x = as.factor(Sex), y = htval)) +
+     geom_boxplot(fill = "lightblue", color = "black") +
+     labs(x = "Gender", y = "Height (cm)", title = "Height by Gender") +
+     scale_x_discrete(labels = c("1" = "Male", "2" = "Female")) +  # Map 1 = Male, 2 = Female
+     theme_minimal()
>
> View(data)
>
>
> # Filter out rows where wtval is non-finite (e.g., NA, Inf)
> data_clean_weight <- data[!is.na(data$wtval) & is.finite(data$wtval), ]
>
> # Boxplot for weight by gender
> ggplot(data_clean_weight, aes(x = as.factor(Sex), y = wtval)) +
+     geom_boxplot(fill = "lightgreen", color = "black") +
+     labs(x = "Gender", y = "Weight (kg)", title = "Weight by Gender") +
+     scale_x_discrete(labels = c("1" = "Male", "2" = "Female")) +  # Map 1 = Male, 2 = Female
+     theme_minimal()
```

Violin plots for T-tests

```
> # Filter out rows where htval is non-finite (e.g., NA, Inf)
> data_clean_height <- data[!is.na(data$htval) & is.finite(data$htval), ]
>
> # Violin plot for height by gender
> ggplot(data_clean_height, aes(x = as.factor(Sex), y = htval)) +
+     geom_violin(fill = "lightblue") +
+     labs(x = "Gender", y = "Height (cm)", title = "Height by Gender") +
+     scale_x_discrete(labels = c("1" = "Male", "2" = "Female")) +  # Map 1 = Male, 2 = Female
+     theme_minimal()
>
> # Filter out rows where wtval is non-finite (e.g., NA, Inf)
> data_clean_weight <- data[!is.na(data$wtval) & is.finite(data$wtval), ]
>
> # Violin plot for weight by gender
> ggplot(data_clean_weight, aes(x = as.factor(Sex), y = wtval)) +
+     geom_violin(fill = "lightgreen") +
+     labs(x = "Gender", y = "Weight (kg)", title = "Weight by Gender") +
+     scale_x_discrete(labels = c("1" = "Male", "2" = "Female")) +  # Map 1 = Male, 2 = Female
+     theme_minimal()
```

Correlation heatmap

```
> library(reshape2)
> library(ggplot2)
>
> # Correlation heatmap
> corr_data <- correlation_matrix
> corr_melt <- melt(corr_data)
>
> ggplot(corr_melt, aes(x = Var1, y = Var2, fill = value)) +
+     geom_tile(color = "white") +
+     scale_fill_gradient2(low = "blue", high = "red", mid = "white", midpoint = 0) +
+     labs(x = "", y = "", title = "Correlation Heatmap") +
+     theme_minimal() +
+     theme(axis.text.x = element_text(angle = 45, hjust = 1))
```